

TECH SPECTRA

2026

HACKATHON

THEME

AI FOR REAL-WORLD PROBLEMS

"Build Solutions That Matter."

OFFICIAL RULEBOOK

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This rulebook is the official governing document for TECH SPECTRA 2026 Hackathon. All participants are required to read and comply with all sections herein.

§ 1

EVENT OVERVIEW

1.1 Introduction

TECH SPECTRA 2026 is the flagship innovation and technology competition organized by Lachoo Memorial College of Science and Technology, Jodhpur. This Hackathon is a dynamic, fast-paced, 2-hour innovation challenge designed to ignite creative thinking and technical problem-solving among student participants.

The event challenges teams of students to identify pressing, genuine real-world problems and develop technology-driven solutions within a defined timeframe. Participants are encouraged to leverage Artificial Intelligence (AI), software development, automation, data analytics, and digital platforms to create meaningful prototypes and proof-of-concept solutions.

1.2 Event Objectives

The TECH SPECTRA 2026 Hackathon is structured to fulfill the following primary objectives:

- **Innovation:** Inspire students to think beyond conventional approaches and develop original, creative solutions to complex challenges.
- **Problem Solving:** Develop participants' capacity to analyze, understand, and address real-world challenges with structured technical thinking.
- **Teamwork & Collaboration:** Foster a spirit of cooperative effort, where diverse skill sets come together to produce cohesive, functional outcomes.
- **Technical Skill Development:** Provide a platform for participants to apply and enhance their technical competencies in programming, AI, and software engineering.
- **AI Utilization:** Encourage the practical and responsible use of Artificial Intelligence tools and methodologies in solving real-world problems.
- **Creativity & Design Thinking:** Cultivate an environment where imaginative ideas are valued alongside technical rigor and academic discipline.

1.3 Hackathon at a Glance

Event Name	TECH SPECTRA 2026 Hackathon
Organized By	Lachoo Memorial College of Science & Technology, Jodhpur
Theme	AI for Real-World Problems
Tagline	"Build Solutions That Matter."
Event Duration	2 Hours (Intensive Innovation Sprint)
Event Mode	On-Campus, In-Person
Participation	Team-Based (2 to 4 Members per Team)

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THEME & PROBLEM DOMAINS

2.1 Official Theme

THEME: AI FOR REAL-WORLD PROBLEMS

TAGLINE: "Build Solutions That Matter."

2.2 Theme Description

The central theme of TECH SPECTRA 2026 challenges participants to move beyond theoretical exercises and engage with authentic, tangible problems faced by communities, institutions, and industries. Participants are required to identify genuine challenges affecting real people and develop technology-powered solutions that have the potential for meaningful, lasting impact.

Solutions must be grounded in real-world applicability. Participants are strongly encouraged to draw from their own experiences, local challenges, or globally recognized issues. The use of Artificial Intelligence as a core component of the solution is highly recommended and will be viewed favorably during the evaluation process.

2.3 Eligible Problem Domains

Participants may choose to address problems from any of the following domains. This list is illustrative and not exhaustive — teams are welcome to explore any domain that presents genuine real-world challenges:

#	Domain	Example Applications
1	Education	EdTech platforms, adaptive learning, student welfare, accessibility tools
2	Healthcare	Medical diagnostics, patient management, mental health, telemedicine
3	Agriculture	Precision farming, crop monitoring, weather prediction, supply chain
4	Smart Cities	Urban planning, waste management, smart infrastructure, citizen services
5	Transportation	Traffic optimization, route planning, public transit, logistics
6	Sustainability & Environment	Climate solutions, renewable energy, pollution monitoring, conservation
7	Student Life	Campus tools, productivity, mental wellness, academic support
8	Business & Commerce	SME tools, e-commerce, financial inclusion, supply chain
9	Public Services	Government services, civic tech, public safety, emergency response
10	Other Domains	Any other domain addressing a genuine, impactful real-world problem

§ 3

ELIGIBILITY CRITERIA

3.1 Who Can Participate?

TECH SPECTRA 2026 is open to all currently enrolled students at Lachoo Memorial College of Science and Technology, Jodhpur, as well as students from other recognized educational institutions, subject to the registration requirements specified below.

3.2 Eligibility Requirements

- **Open Participation:** The event is open to all bonafide students currently enrolled in any undergraduate, postgraduate, or diploma programme at a recognized college or university.
- **Valid College ID:** All participants must carry their valid, original College/University Student Identification Card at the time of registration and throughout the event. Failure to produce a valid ID may result in disqualification.
- **Pre-Registration Mandatory:** Registration must be completed before the event. Walk-in participation on the day of the event without prior registration will not be permitted under any circumstances.
- **Team Registration:** All participants must register as part of a team. Individual registrations are not accepted. Refer to Section 4 for team composition rules.
- **Single Team Membership:** A student may be a member of only one registered team. Participation in multiple teams is strictly prohibited.
- **Honest Representation:** All information provided during registration must be accurate and truthful. Providing false information shall lead to immediate disqualification.

§ 4

TEAM COMPOSITION

4.1 Team Size Requirements

Parameter	Requirement
Minimum Team Size	2 Members
Maximum Team Size	4 Members
Team Leader	Each team must designate one Team Leader who will act as the primary point of c
Inter-Department Teams	Teams comprising students from different departments or courses are permitted a
External Collaboration	All team members must be registered students. External collaborators or non-stud

4.2 Team Name & Identity

Each team must register under a unique Team Name. The Team Name must be appropriate, professional, and must not contain offensive, obscene, or politically sensitive content. The organizing committee reserves the right to request a change of Team Name if deemed inappropriate.

Teams are encouraged to choose creative names that reflect their identity and the spirit of innovation. The Team Name registered at the time of sign-up will be used throughout the event, including during presentations and awards.

§ 5

HACKATHON FORMAT & SCHEDULE

5.1 Total Duration

Total Event Duration: 2 Hours (Intensive Hackathon Sprint)
Format: In-Person, On-Campus Competition
All phases are time-bound. Strict adherence to timelines is mandatory.

5.2 Event Flow & Phases

Phase	Stage	Description
Phase 1	Theme Briefing & Orientation	Official commencement of the hackathon. The organizing committee will formally announce the theme, rules, and objectives.
Phase 2	Development Phase	The core innovation sprint. Teams work collaboratively to define their problem statement, brainstorm ideas, and develop prototypes.
Phase 3	Submission	Teams must formally submit their project documentation, including the mandatory Poster/Presentation Slides, by the deadline.
Phase 4	Presentation	Each team presents their solution to the panel of judges in a structured format: 5 minutes for presentation + 5 minutes for Q&A.
Phase 5	Judging & Evaluation	The panel of judges evaluates all presented solutions based on the official Judging Criteria (Innovation, Feasibility, Impact, etc.).
Phase 6	Results & Closing	Official announcement of results, recognition of winning teams, and formal closing of the event.

§ 6

ALLOWED TECHNOLOGIES

6.1 Technology Freedom Policy

TECH SPECTRA 2026 is a technology-agnostic hackathon. Participants are free to use any programming language, framework, tool, or platform of their choice. The organizing committee does not impose restrictions on the technology stack used, provided that the tools are legally accessible, free to use during the event, and do not violate any intellectual property rights.

6.2 Permitted Technology Categories

Category	Examples & Details
Programming Languages	Python, JavaScript, Java, C++, C#, TypeScript, Go, Rust, PHP, Swift, Kotlin, R, MATLAB, and
Frameworks & Libraries	React, Angular, Vue.js, Node.js, Django, Flask, FastAPI, Spring Boot, .NET, TensorFlow, PyT
AI & ML Tools	All AI-powered tools and platforms including large language models, computer vision APIs, N
APIs & Web Services	Any publicly available REST APIs, GraphQL APIs, cloud service APIs, or third-party data serv
Development Platforms	GitHub, GitLab, Replit, Google Colab, Jupyter Notebooks, VS Code, and any standard develo
Cloud Services	AWS, Google Cloud, Microsoft Azure, Firebase, Supabase, and any cloud platform with a free
Design & Prototyping	Figma, Canva, Adobe XD, or any other design tools for UI/UX mockups and prototyping

6.3 Notable Permitted AI Tools (Examples)

The following are examples of AI tools that participants may utilize. This list is not exhaustive:

- ChatGPT (OpenAI) — Text generation, code assistance, brainstorming
- Claude (Anthropic) — Reasoning, writing, analysis, code generation
- Gemini (Google) — Multimodal AI, research, content creation
- GitHub Copilot — AI-powered code completion and suggestion
- Perplexity AI — AI-powered research and search
- Midjourney / DALL-E — AI image generation for UI/UX design concepts
- Hugging Face Models — Open-source NLP, vision, and audio models

§ 7

AI USAGE POLICY

7.1 Policy Statement

TECH SPECTRA 2026 acknowledges that Artificial Intelligence tools are an integral part of modern software development and problem-solving. Accordingly, the use of AI tools is explicitly permitted during the hackathon. Participants are encouraged to leverage AI responsibly and intelligently to enhance the quality, speed, and impact of their solutions.

7.2 Conditions for AI Use

While AI tool usage is unrestricted, the following conditions must be strictly adhered to:

- **Full Understanding Required:** Participants must thoroughly understand every component of their submitted solution. The use of AI to generate code, content, or designs does not exempt a team from the responsibility of understanding what has been built. Presenting work that the team cannot explain will be viewed negatively by judges.
- **Explainability During Q&A:** During the Q&A segment of the presentation (Section 10), judges are entitled to ask detailed technical questions regarding the architecture, logic, workflow, and implementation of the solution. Teams must be prepared to answer such questions with clarity and confidence.
- **Architecture & Workflow Knowledge:** Teams must be capable of explaining the overall system architecture, data flow, AI model selection, API integrations, and the rationale behind every major design decision made in the project.
- **Impact on Scoring:** A team's inability to satisfactorily explain their solution, even if technically impressive, will adversely affect their score under the Technical Implementation and Presentation & Demo judging criteria (refer Section 11).
- **AI-Generated Content Disclosure:** Teams that use AI-generated code, text, or graphics as a substantial component of their solution are encouraged to disclose this during their presentation. Transparency will be valued positively.
- **No Outsourcing:** AI tools may be used as assistants, not as outsourced developers. The intellectual direction, problem identification, solution design, and creative decisions must originate from the team members themselves.

§ 8

PROHIBITED ACTIVITIES

8.1 Zero Tolerance Policy

TECH SPECTRA 2026 maintains a strict zero-tolerance policy towards dishonesty, misconduct, and any form of unethical behavior. Violations of the prohibited activities listed herein will result in immediate disqualification and may be reported to the appropriate college authorities.

8.2 List of Prohibited Activities

- 1. Plagiarism:** Submitting work that is copied, reproduced, or derived from another team's project, existing open-source projects, online repositories, or any prior hackathon submissions without substantial original contribution.
- 2. Copying Another Team's Work:** Directly or indirectly replicating another team's solution, problem statement, code, design, or presentation during the event.
- 3. Purchasing or Outsourcing Projects:** Acquiring a pre-built project, code, or complete solution from any commercial service, freelancer, or online marketplace and presenting it as original work.
- 4. Misconduct & Disruptive Behavior:** Any behavior that disrupts the event proceedings, interferes with other teams' work, creates a hostile environment, or violates the dignity of fellow participants, judges, or organizers.
- 5. Impersonation:** Registering or participating under another student's name, college ID, or identity. Allowing a non-registered individual to participate on behalf of a registered team member.
- 6. Pre-Built Solutions:** Submitting projects that were fully built or substantially completed before the hackathon commenced. All work must be produced during the designated development phase.
- 7. Unauthorized Resource Access:** Accessing, copying, or interfering with another team's files, code repositories, or project materials without permission.
- 8. Violation of Rulebook:** Any action that contravenes the rules, guidelines, and policies outlined in this official rulebook.
- 9. Unethical Use of AI:** Using AI tools to fabricate data, generate false research findings, impersonate subject matter experts, or create deliberately misleading content.

§ 9

SUBMISSION REQUIREMENTS

9.1 Submission Overview

Every registered team is required to make a formal submission before the designated submission deadline. Incomplete submissions or submissions made after the cutoff time will not be accepted and may result in the team being ineligible for evaluation.

9.2 Mandatory Submission Components

#	Submission Component	Details & Requirements
1	Team Name	The officially registered team name as submitted during registration
2	Project Name	A clear, descriptive name for the solution developed during the hackathon
3	Team Members List	Full names of all team members with their student roll numbers and department/course
4	Problem Statement	A concise written statement (minimum 100 words) describing the real-world problem being
5	PPT Presentation	A PowerPoint presentation (MANDATORY) covering: Problem Statement, Proposed Solution
6	Working Demo / Prototype	A live working demonstration or functional prototype of the solution. If a fully working demo

9.3 Submission Guidelines

- All submissions must be made to the designated volunteer or submission station before the official deadline.
- The Team Leader is responsible for ensuring the completeness and accuracy of the submission.
- Submissions must be original work produced during the event. Prior work or pre-built components must be explicitly disclosed.
- Teams should retain a personal backup of all submitted materials.
- Any submission found to be incomplete may be rejected at the discretion of the organizing committee.

§ 10

PRESENTATION RULES

10.1 Presentation Format

Presentation Time: 5 Minutes per Team (Strict Time Limit)

Q&A Session: 2 Minutes (Judges' Questions)

Total Slot: 7 Minutes per Team

WARNING: Teams exceeding the time limit will be stopped immediately.

10.2 Presentation Requirements

All team members are encouraged to participate actively in the presentation. The use of the submitted PowerPoint presentation is mandatory. Teams may use live demonstrations, video clips (not exceeding 90 seconds), or interactive prototypes to supplement their presentation.

10.3 Recommended Presentation Structure

Slide	Content	Recommended Time
1	Title Slide — Team Name, Project Name, Team Members	30 seconds
2	Problem Statement — What problem? Who is affected? Why does it matter?	60 seconds
3	Proposed Solution — How does your solution address the problem?	60 seconds
4	Technology Stack & Architecture — Tools, frameworks, AI used	45 seconds
5	Live Demo / Prototype Walkthrough — Show your working solution	75 seconds
6	Impact & Scalability — Real-world potential, future scope	30 seconds

10.4 Additional Presentation Rules

- Teams must be present and ready at their designated presentation slot. Failure to be present when called may result in forfeiture of the presentation opportunity.
- Presentations must be conducted in English or Hindi. Mixing of languages is acceptable.
- Props, physical prototypes, or hardware demonstrations are permitted provided they do not disrupt other teams or the venue.
- All team members must be present during the presentation. Absent members may not present remotely.
- Judges reserve the right to ask questions at any point during the presentation and Q&A.;
- Disrespectful or unprofessional conduct during another team's presentation will be penalized.

§ 11

JUDGING CRITERIA & SCORING

11.1 Evaluation Framework

All submitted solutions will be evaluated by a qualified panel of judges comprising faculty members, industry professionals, and/or domain experts appointed by the organizing committee. Judges will assess each team's solution independently based on the following official judging criteria. The panel's decision shall be final and binding.

11.2 Official Scoring Rubric

#	Judging Criterion	Max Marks	What Judges Evaluate
1	Problem Identification	20	Clarity and relevance of the problem statement; evidence of genuine real-world need
2	Innovation & Creativity	25	Originality of the approach; unique application of technology; creative thinking in solution
3	Technical Implementation	25	Quality of code/prototype; correct use of technology; AI integration; system architecture
4	Impact & Scalability	20	Potential real-world impact; feasibility of deployment; scalability to larger audiences
5	Presentation & Demo	10	Clarity of communication; quality of slides; effectiveness of demo; team's ability to answer questions
TOTAL SCORE		100	Combined evaluation across all five criteria

11.3 Scoring Notes

- Judges may independently assign scores; final scores will be averaged across the panel.
- In the event of a tie, judges will conduct a brief tiebreaker Q&A; with the tied teams.
- Teams that fail to submit the mandatory PPT presentation will be disqualified from evaluation.
- Teams that cannot explain their solution during Q&A; may have their Technical Implementation score reduced by up to 50%.
- Bonus consideration may be given to solutions demonstrating exceptional social impact or innovative AI application, at the judges' discretion.

§ 12

DISQUALIFICATION POLICY

12.1 Grounds for Disqualification

The organizing committee and judging panel reserve the absolute right to disqualify any team or individual participant from TECH SPECTRA 2026 if any of the following grounds are established. Disqualification may occur at any stage of the event, including after results have been declared, if violations are discovered subsequently.

X Plagiarism: Submission of work that is copied, stolen, or derived from pre-existing projects, another team's work, or any source not properly attributed.

X Academic or Professional Misconduct: Any behavior that violates academic integrity standards, including falsifying data, misrepresenting capabilities, or making false claims about the solution.

X Provision of False Information: Providing inaccurate, misleading, or fabricated information during registration, submission, or presentation.

X Violation of Event Rules: Breaching any rule, guideline, or policy outlined in this official rulebook.

X Unethical Behavior: Engaging in any conduct that is dishonest, harmful, disrespectful, or that brings the event, college, or fellow participants into disrepute.

X Impersonation or Identity Fraud: Participating under a false identity or allowing unauthorized individuals to participate in place of registered team members.

X Submission of Pre-Built Work: Submitting a project that was substantially completed prior to the commencement of the hackathon's development phase.

12.2 Disqualification Procedure

The organizing committee will follow a fair and transparent process for disqualification. If a violation is suspected, the concerned team will be notified and given an opportunity to respond. The final decision regarding disqualification rests solely with the organizing committee, and their decision shall be final and binding.

§ 13

ORGANIZING COMMITTEE DECISION

13.1 Authority & Finality

TECH SPECTRA 2026 is organized and governed by the Event Organizing Committee of Lachoo Memorial College of Science and Technology, Jodhpur. The Organizing Committee has been authorized by the college administration to manage all aspects of the event, including but not limited to: registration, rule interpretation, judging oversight, dispute resolution, and results declaration.

IMPORTANT: The decision of the Organizing Committee and the Panel of Judges shall be FINAL and BINDING in all matters related to TECH SPECTRA 2026.
No appeals, objections, or grievances will be entertained after the official announcement of results.

13.2 Scope of Authority

- **Rule Interpretation:** The Organizing Committee holds exclusive authority to interpret any ambiguous rule or guideline. Their interpretation shall be considered definitive and binding upon all participants.
- **Dispute Resolution:** Any disputes arising between teams, or between a team and the organizing committee, shall be resolved by the Organizing Committee. The resolution process will be fair but the final decision shall not be subject to appeal.
- **Event Modifications:** The Organizing Committee reserves the right to modify event timelines, rules, or procedures during the event if circumstances necessitate such changes, with immediate effect and notification to all participants.
- **Results Declaration:** The declaration of results by the Organizing Committee, in consultation with the Panel of Judges, shall be considered final. No post-declaration appeals regarding scoring or ranking will be accepted.
- **Extraordinary Circumstances:** In case of any situation not covered by this rulebook, the Organizing Committee shall exercise its judgment to arrive at the most fair and appropriate decision.

§ 14

CODE OF CONDUCT**14.1 Preamble**

TECH SPECTRA 2026 is committed to providing a safe, respectful, inclusive, and productive environment for all participants, organizers, judges, volunteers, and observers. Every individual associated with this event is expected to uphold the highest standards of personal and professional conduct. This Code of Conduct applies throughout all phases of the event, including registration, development, presentation, and post-event activities.

14.2 Expected Conduct — All Participants**■ Respect & Courtesy**

Treat all fellow participants, organizers, judges, and volunteers with dignity and respect. Discriminatory, offensive, or derogatory language or behavior based on gender, caste, religion, nationality, disability, or any other characteristic is strictly prohibited and will result in immediate removal from the event.

■ Intellectual Honesty

All submitted work must represent the team's original, authentic effort. Representing others' work as your own, whether from another team, the internet, or AI without understanding, is a violation of intellectual honesty and the foundational principles of this event.

■ Collaborative Spirit

While TECH SPECTRA 2026 is a competition, participants are encouraged to maintain a spirit of healthy sportsmanship. Celebrate the achievements of others, offer support where appropriate, and compete with grace and fairness.

■ Professional Communication

All communications — verbal and written — during the event must be professional, constructive, and respectful. Aggressive, threatening, or inflammatory communication will not be tolerated.

■ Venue & Equipment Responsibility

Participants are responsible for the careful use of all venue facilities, equipment, and resources. Any damage caused by a participant or team will be their sole responsibility. Food, beverages, and personal items must be managed responsibly.

■ Noise & Disruption

Teams must maintain reasonable noise levels within the workspace. Excessive noise, disruptive behavior, or activities that hinder other participants' ability to work are prohibited.

■ Privacy & Confidentiality

Participants must not access, copy, photograph, or interfere with another team's work, materials, or devices under any circumstances. Respect for intellectual property and personal privacy is mandatory.

■ Device & Technology Use

All electronic devices — laptops, phones, tablets — must be used solely for hackathon-related activities during the development phase. Use of devices for purposes unrelated to the event during work time is discouraged.

■ Reporting Violations

If a participant witnesses any form of misconduct, harassment, or violation of these rules, they are encouraged to report the matter immediately to a designated event volunteer or organizing committee member. All reports will be treated confidentially and addressed promptly.

■ Compliance with Organizer Instructions

All participants must comply with the reasonable instructions given by organizers, volunteers, security personnel, and judges during the event. Failure to comply with official instructions may result in removal from the event premises.

■ Social Media Conduct

Participants are welcome to share their hackathon experience on social media. However, they must not post content that misrepresents the event, defames other participants, or violates the privacy of others. Use of the official event hashtag is encouraged.

■ Dress Code

Participants are expected to maintain a decent, appropriate dress code befitting a professional academic environment. Offensive or inappropriate attire will not be permitted.

14.3 Consequences of Code Violations

Violations of this Code of Conduct will be addressed swiftly and fairly by the Organizing Committee. Depending on the severity of the violation, consequences may include: a formal warning, reduction in score, disqualification from the competition, removal from the event premises, and/or reporting to the college administration for further disciplinary action. The safety and wellbeing of all participants is the organizing committee's highest priority.

THEME: AI FOR REAL-WORLD PROBLEMS

"Build Solutions That Matter."

- This Official Rulebook has been prepared for TECH SPECTRA 2026 Hackathon, organized by the Department of Computer Science & Engineering, Lachoo Memorial College of Science and Technology, Jodhpur.
- All participants are required to have read, understood, and agreed to all rules and guidelines contained herein prior to participation.
- The Organizing Committee reserves the right to amend, update, or clarify any rule contained in this document. Participants will be notified of any changes through official event channels.
- TECH SPECTRA 2026 is an official college academic event. All participants are expected to uphold the values and reputation of Lachoo Memorial College of Science & Technology.

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